

KeepingCool_Analysis (v1.0.1)

Repository: https://github.com/Baileys42/KeepingCool_Analysis/tree/v1.0.1

This repository contains the analysis pipeline used to process two-colour single-molecule FRET (smFRET) data: importing idealized HMM-fitted traces (generated by vbFRET), removing outliers, and generating FRET efficiency histograms, synchronised-transition heatmaps, transition density plots (TDPs), and residence time/kinetics analyses.

System Requirements

Software dependencies

The two-colour FRET pipeline (`Experiment_1/src/two-colour-FRET/`) requires Python 3 and the following packages. Versions below match those in the provided `twocolourenvironment.yml`:

Package	Version
Python	3.8.13
pandas	1.2.3
numpy	1.19.2
matplotlib	3.5.1
seaborn	0.11.1
scipy	1.6.1
lmfit	1.0.3
scikit-learn	(used for Gaussian Mixture Model fitting in <code>gaussian_plots.py</code>)

Standard library modules (`os`, `glob`, `re`, `shutil`, `math`, `functools`, `warnings`, `itertools`) are also used and require no separate installation.

Operating systems

Tested on Windows 10/11 (64-bit). The code is pure Python and should run on macOS/Linux, but this has not been tested.

Hardware requirements

No non-standard hardware (e.g., GPU) is required. A standard desktop or laptop is sufficient.

Installation Guide

1. Clone the repository (v1.0.1 tag):

```
git clone --branch v1.0.1 https://github.com/Baileys42/KeepingCool_Analysis.git
cd KeepingCool_Analysis
```

2. Create the analysis environment from the provided two-colour environment file:

```
conda env create -f twocoloureenvironment.yml
conda activate sm_analysis_3
```

Typical install time

Cloning the repository takes a few seconds. Creating the conda environment typically takes 5–15 minutes, depending on internet connection speed.

Demo

This demo walks through generating a FRET efficiency histogram from idealized smFRET traces.

Step 1: Fill out Example_data_file.csv with treatment labels and folder paths

Open Experiment_1/raw_data/Example_data_file.csv:

```
Treatment,Repeat,Date,Directory
treat_name,1,121212,path/to/folder/
treat_name,2,121213,path/to/folder/
```

For each row: - **Treatment**: a label for the experimental condition (e.g., DnaK, Control). Rows sharing the same **Treatment** are combined as technical repeats. - **Repeat**: the technical repeat number for that condition. - **Date**: the date the data was collected (for record-keeping only). - **Directory**: the path to the folder containing the idealized .dat trace files exported from vbFRET for that treatment/repeat. The path must end with a trailing slash (/).

Add additional rows for each treatment and repeat to be included in the analysis.

Step 2: Run lines 1–2 and 9–11 of 0-import_data.py

```
from smfret.src.processing_scripts import data_import_script as ps
import pandas as pd

output_folder = 'Experiment_1/python_results'
data = pd.read_csv('Experiment_1/raw_data/Example_data_file.csv')
compiled_df = ps.combine_technical_repeats(data, output_folder)
```

This reads Example_data_file.csv, imports the .dat files from each Directory, removes outlier FRET values, and concatenates everything into a single dataframe. Two output files are written to Experiment_1/python_results/:

- Cleaned_FRET_histogram_data.csv
- sep_repeat.csv

Step 3: Run all of 1-histogram_timelapse_traces_synchtrans_heatmap.py

This script reads Cleaned_FRET_histogram_data.csv (generated in Step 2) and runs the following:

- `pa.master_histogram_func(...)` — plots FRET efficiency histograms (ridgeline and bar) and the proportion of time each molecule spends below the FRET threshold (`FRET_thresh = 0.5`).

- `ph.plot_gmm_fits(...)` — fits a Gaussian Mixture Model to each treatment’s FRET distribution.
- `pg.master_plot_synchronised_transitions(...)` — identifies and plots synchronised FRET transitions, generating heatmap dataframes.
- `pe.plot_fret_transition_kde(...)` — generates a KDE heatmap of FRET behaviour around transitions.

Step 4: Expected output

The primary output of this demo is a FRET efficiency histogram, saved to:

- `Experiment_1/python_results/Histogram_bar.svg`
- `Experiment_1/python_results/Histogram-ridgeline.svg`

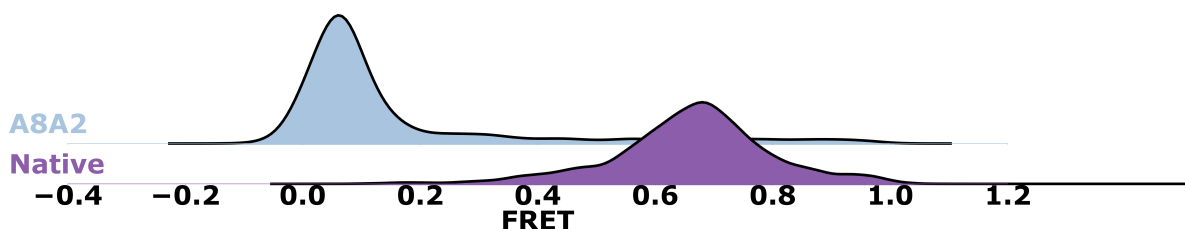


Figure 1: *[Example ridgeline result from provided data]*

Step 5: Optional kinetics analysis

Running `2-transition-and-residence-time-analysis.py` performs additional kinetic analyses on the data processed in Steps 2–3:

- `pi.master_tdp_cleanup_func(...)` and `pi.master_TDP_plot(...)` — generate transition density plots (TDPs).
- `pj.master_dwell_time_func(...)` — plots dwell time data (e.g., bind-and-release events).
- `pk.transition_frequency_plot(...)` — plots transition frequency data.
- `pl.master_residence_time_func(...)` — plots residence time data (mean $\pm$ SE, violin plots, and cumulative histograms with single/double exponential fits).

Instructions for Use

To run the pipeline on your own data:

1. Export idealized traces for each molecule from vbFRET as individual `.dat` files, organised into one folder per treatment/repeat.
2. Create a copy of `Example_data_file.csv` and add one row per treatment/repeat, with `Treatment` set to a meaningful condition name and `Directory` pointing to the corresponding folder of `.dat` files.
3. Update `output_folder` in `0-import_data.py`, `1-histogram_timelapse_traces_synchtrans_heatmap.py` and `2-transition-and-residence-time-analysis.py` to point to your desired results directory.
4. Run lines 1–2 and 9–11 of `0-import_data.py` to import and clean your data.
5. Run `1-histogram_timelapse_traces_synchtrans_heatmap.py` to generate FRET histograms, GMM fits, and synchronised transition heatmaps.
6. Adjust analysis parameters as needed for your dataset, e.g.:
 - `FRET_thresh` (FRET threshold used to define the bound/unbound or low/high FRET states)
 - `exposure` (camera exposure time, in seconds)
 - `frames_to_plot`, `FRET_before`, `FRET_after` (synchronised transition window settings)
7. Optionally run `2-transition-and-residence-time-analysis.py` for TDP, dwell time, and residence time analyses, adjusting `transition_threshold`, `fps`, and `thresh` to match your acquisition settings.

Reproduction Instructions

To reproduce the FRET efficiency histograms and kinetic analyses reported in the manuscript:

1. Obtain the raw idealized `.dat` trace files associated with the manuscript (see the manuscript’s Data Availability statement).
2. Populate a data file (in the same format as `Example_data_file.csv`) with one row per treatment and technical repeat, pointing `Directory` to the corresponding folder of `.dat` files.
3. Run the pipeline as described in the Demo above (Steps 2–5), using the same parameter values as in the provided scripts (`FRET_thresh` = 0.5, `exposure` = 0.2, `frames_to_plot` = 50, `FRET_before` = `FRET_after` = 0.5, `transition_threshold` = 0.5, `fps` = 5).

4. Output figures (histograms, GMM fits, synchronised transition heatmaps, TDPs, dwell time, and residence time plots) will be written to the specified `output_folder`.